
Training Artemis Astronauts to Explore the Moon

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Abstract Text:

To prepare for human exploration of the Moon in the mid-2020s, NASA is developing a program to provide geoscience and planetary science classroom training and geologic field training to astronauts and other mission personnel. The Artemis geoscience training plan builds on a long legacy: NASA has trained astronauts in geology, planetary science, and geological field work for more than 50 years. From ~1965-1972, the geology focus for astronauts included orbital Earth observations during the early NASA missions, but the endgame was preparation for the Apollo missions to the Lunar surface [see Phinney (2015), NASA-SP-2015-626; Lofgren et al.(2011); Schmitt, et al. (2011); and El Baz (2011), all from GSA SP483]. From the mid 1970s through 2017, the primary purpose for geoscience training was to prepare astronauts to observe the Earth and document a wide variety of Earth processes from orbital platforms (Skylab, Space Shuttle and International Space Station). Astronaut observations of Earth from the International Space Station remain an important activity today, but the prospect of exploration of the lunar surface through the Artemis Program lends additional purpose to the geoscience content and field experiences that we provide the astronauts. We recently completed the geology and field training with Astronaut Class 22 and are now planning both focused field exercises and simulations for future Artemis crews, as well as the accompanying classroom lessons on lunar and planetary sciences. The training we offer extends beyond the astronaut office – we partner across operational and engineering organizations at NASA and provide fieldwork exercises and opportunities to personnel in the Flight Operations Directorate (responsible for the overall training and certification), as well as the exploration Extravehicular Activity Office (developing spacewalk suits and tools). These partnerships contribute to building a common language and sense of purpose, and also help to ensure that NASA's geoscience training and science operations are fully integrated with the teams and systems responsible for getting the next generation of astronauts to the surface of the Moon.

Plain-Language Summary:

NASA scientists are developing a training program in geology and planetary science for astronauts going to explore the Moon. The program includes both classroom lessons and training in different field locations. We also provide field training to NASA engineers developing the plans for lunar missions and building mission hardware like spacesuits. Our goal is to contribute to mission

success by training different groups at NASA (astronauts, engineers, mission operations managers) how to think and work like field geologists, and to better understand the important science questions that we hope to answer by sending humans to the Moon.

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